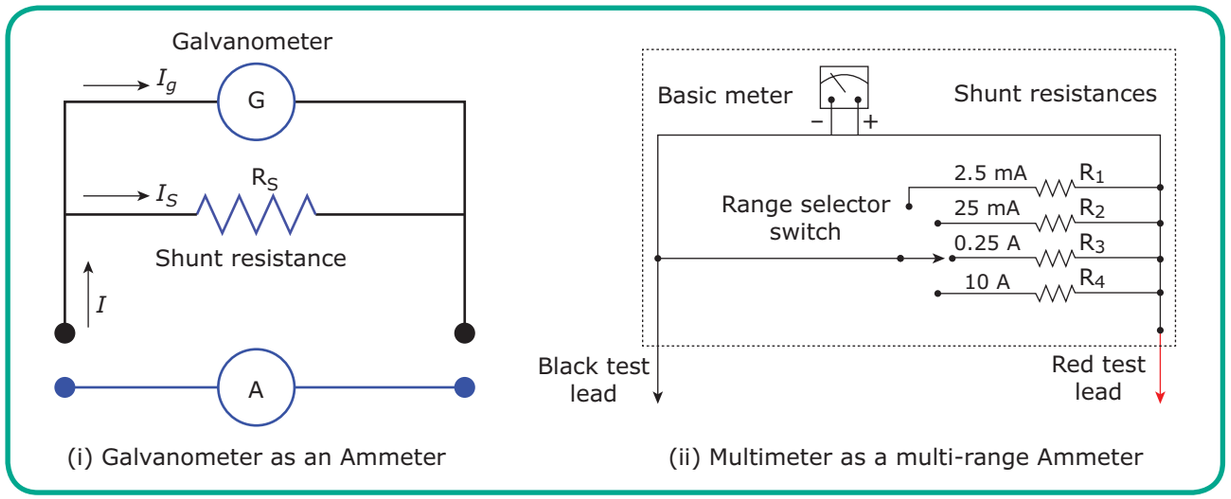


Figure 10.7 **Multimeter as an Ammeter - circuit**

to provide a required number of current ranges as shown in Figure 10.7. To measure AC current, a diode should be connected in series with the meter.

10.2.2.4. Multimeter as an Ohmmeter

An instrument which is used for measuring ohms is called as Ohm meter.

Figure 10.8 shows the circuit of an Ohmmeter. A fixed resistance R and a variable resistance R_V are connected in series with the meter and a battery (fixed inside the meter). The fixed resistance R limits the current within the desired range and the variable resistance R_V is used for setting zero ohm of the needle before taking any range of resistance measurements. This setting is called as zero ohm adjustment. As a result, accurate ohmic readings can be measured.

Two probes (A and B terminals) of the multimeter are connected across the resistance to be measured. If the resistance is high, lower current flows through the circuit and the meter will indicate higher ohms and vice versa.

To measure multi range Ohmic readings, various resistance values such as R_1, R_2 & R_3 should be connected in series with battery through selector switch as shown in the Figure 10.8 (ii)

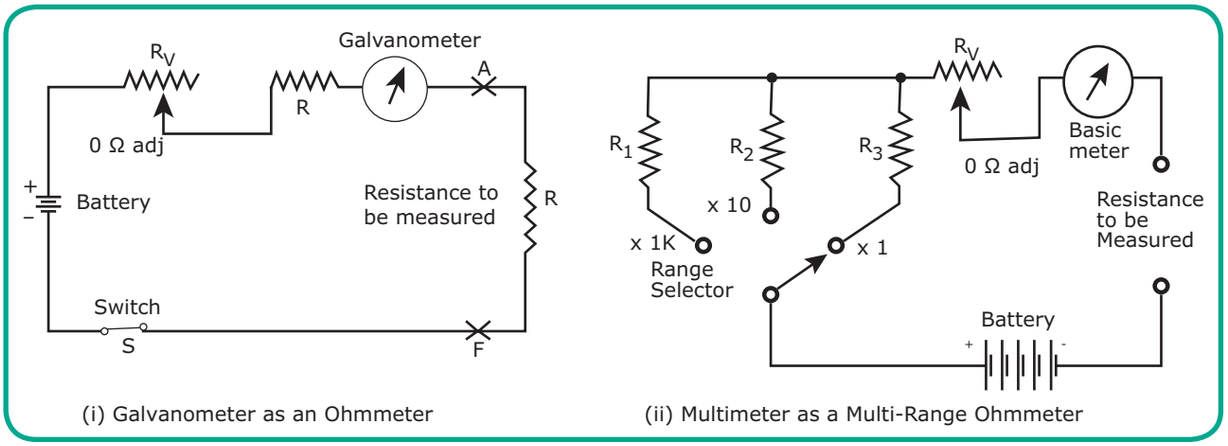


Figure 10.8 **Multimeter as an Ohmmeter - circuit**

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